

Description

The μPD41416 is a 16,384-word by 4-bit dynamic N-channel MOS RAM designed to operate from a single +5 V power supply. The negative voltage substrate bias is internally generated; its operation is both automatic and transparent. The μPD41416 utilizes a double-polylayer, N-channel, silicon gate process which provides high storage cell density, high performance, and high reliability.

The μPD41416 uses a single transistor dynamic storage cell and advanced dynamic circuitry throughout, ensuring minimum power dissipation. Refresh is accomplished by performing RAS-only refresh cycles, hidden refresh cycles, or normal read or write cycles on the 128 address combinations of A₀-A₆ during the refresh period of 2 milliseconds.

Multiplexed address inputs permit the μPD41416 to be packaged in a standard 18-pin dual-in-line package for high system bit density.

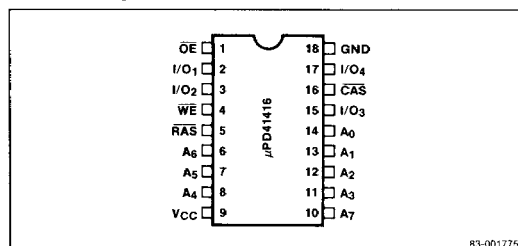
Features

- ☐ 16,384-word × 4-bit organization
- ☐ Single +5 V power supply ± 10%
- ☐ Standard 18-pin plastic package
- ☐ CAS, $\overline{OE}$ or early write mode to control D_{OUT} buffer impedance
- ☐ Low power dissipation,
 - Active (t_{RC} = min): 303 mW
 - Standby: 28 mW
- ☐ Read, write, read-write, read-modify-write. $\overline{RAS}$ -only refresh, hidden refresh, and page mode capabilities
- ☐ 128 refresh cycles during 2 ms period

Performance Ranges

Device	t _{RAC}	t _{CAC}	t _{OEA}
μPD41416-12	120 ns	60 ns	30 ns
μPD41416-15	150 ns	75 ns	40 ns
μPD41416-20	200 ns	100 ns	50 ns

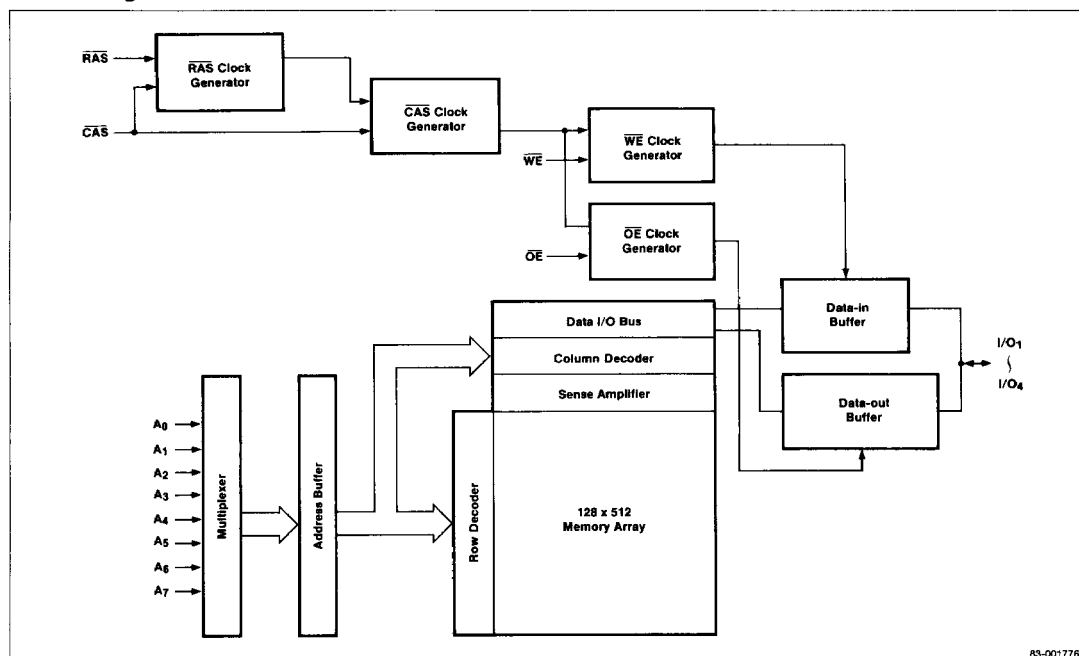
Pin Configuration



Pin Identification

No.	Symbol	Function
1	$\overline{OE}$	Output enable
2-3, 15, 17	I / O ₁ -I / O ₄	Data input / output
4	$\overline{WE}$	Write enable
5	$\overline{RAS}$	Row address strobe
6-8, 10-14	A ₀ -A ₇	Address inputs: A ₀ -A ₅ = Column address inputs A ₀ -A ₆ = Refresh address A ₀ -A ₇ = Row address inputs
9	V _{CC}	+5 V power supply
16	$\overline{CAS}$	Column address strobe
18	GND	Ground

Block Diagram



Absolute Maximum Ratings

Voltage on any pin relative to GND	-1.0 V to +7.0 V
Storage temperature, T_{STG}	-55°C to 125°C
Short-circuit output current, I_{OS}	50 mA
Power dissipation, P_D	1.0 W

Comment: Exposing the device to stresses above those listed in Absolute Maximum Ratings could cause permanent damage. The device is not meant to be operated under conditions outside the limits described in the operational sections of the specification. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

Capacitance

$T_A = 0^\circ\text{C}$ to $+70^\circ\text{C}$, $V_{CC} = 5.0\text{ V} \pm 10\%$; $f = 1.0\text{ MHz}$

Parameter	Symbol	Limits			Unit	Test Conditions
		Min	Typ	Max		
Input capacitance, address inputs	C_{I1}			5	pF	
Input capacitance, strobe inputs	C_{I2}			8	pF	
Input/output capacitance, data ports	$C_{I/O}$			7	pF	

DC Characteristics

$T_A = 0^\circ\text{C}$ to $+70^\circ\text{C}$

Parameter	Symbol	Limits			Unit	Test Conditions
		Min	Typ	Max		
Supply voltage, high	V_{CC}	4.5	5.0	5.5	V	
Supply voltage, low	GND	0	0	0	V	
Standby supply current	I_{CC2}			5.0	mA	$\overline{\text{RAS}} = V_{IH}$, $D_{OUT} = \text{High impedance}$
Input leakage current	$I_{I(L)}$	-10		10	μA	$0\text{ V} \leq V_{IN} \leq V_{CC}$, all other pins not under test = 0 V
Output leakage current	$I_{O(L)}$	-10		10	μA	D_{OUT} is disabled, $0\text{ V} \leq V_{OUT} \leq +5.5\text{ V}$
Output voltage, low	V_{OL}	0		0.4	V	$I_{OUT} = 4.2\text{ mA}$
Output voltage, high	V_{OH}	2.4			V	$I_{OUT} = -2\text{ mA}$
Input voltage, low	V_{IL}	-1.0		0.8	V	
Input voltage, high	V_{IH}	2.4		5.5	V	

AC Characteristics (Notes 2, 3, 4)

$T_A = 0^\circ\text{C to } +70^\circ\text{C}$, $V_{CC} = 5.0\text{V} \pm 10\%$ (Note 1)

Parameter	Symbol	Limits						Unit	Test Conditions
		μPD41416-12		μPD41416-15		μPD41416-20			
		Min	Max	Min	Max	Min	Max		
Operating supply current, average	I _{CC1}		55		50		45	mA	RAS, CAS cycling, t _{RC} = t _{RC min.} (Note 5)
Operating supply current, refresh mode, average	I _{CC3}		45		40		35	mA	RAS cycling, CAS = V _{IH} , t _{RC} = t _{RC min.} (Note 5)
Operating supply current, page mode operation, average	I _{CC4}		45		40		35	mA	RAS = V _{IL} , CAS cycling, t _{PC} = t _{PC min.} (Note 5)
Random read or write cycle time	t _{RC}	220		260		330		ns	(Note 6)
Read-write cycle time	t _{RWC}	300		355		445		ns	(Note 6)
Page mode cycle time	t _{PC}	120		145		180		ns	(Note 6)
Access time from RAS	t _{RAC}		120		150		200	ns	(Notes 7, 8)
Access time from CAS	t _{CAC}		60		75		100	ns	(Notes 7, 9)
Output turn-off delay from CAS	t _{OFF}	0	30	0	40	0	50	ns	(Note 10)
Transition time, rise and fall	t _T	3	50	3	50	3	50	ns	(Note 4)
RAS precharge time	t _{RP}		90		100		120	ns	
RAS pulse width	t _{RAS}	120	10,000	150	10,000	200	10,000	ns	
RAS hold time	t _{RSH}	60		75		100		ns	
CAS pulse width	t _{CAS}	60	10,000	75	10,000	100	10,000	ns	
CAS hold time	t _{CSH}	120		150		200		ns	
RAS to CAS delay time	t _{RCD}	25	60	25	75	30	100	ns	(Note 11)
CAS to RAS precharge time	t _{CRP}	0		0		0		ns	(Note 12)
CAS precharge time, non-page cycle	t _{CPN}	25		25		30		ns	
CAS precharge time, page cycle	t _{CP}	50		60		70		ns	
RAS precharge, CAS hold time	t _{RPC}	0		0		0		ns	
Row address setup time	t _{ASR}	0		0		0		ns	
Row address hold time	t _{RAH}	15		15		20		ns	
Column address setup time	t _{ASC}	0		0		0		ns	
Column address hold time	t _{CAH}	20		25		30		ns	
Column address hold time referenced to RAS	t _{AR}	80		100		130		ns	
Read command setup time	t _{RCS}	0		0		0		ns	
Read command hold time referenced to RAS	t _{RRH}	20		20		20		ns	(Note 13)
Read command hold time referenced to CAS	t _{RCH}	0		0		0		ns	(Note 13)
Write command hold time	t _{WCH}	35		45		55		ns	
Write command hold time referenced to RAS	t _{WCR}	95		120		155		ns	
Write command pulse width	t _{WP}	35		45		55		ns	
Write command to RAS lead time	t _{RWL}	40		45		55		ns	
Write command to CAS lead time	t _{CWL}	40		45		55		ns	
Data-in setup time	t _{DS}	0		0		0		ns	(Note 14)

AC Characteristics (Notes 2, 3, 4) (cont) $T_A = 0^\circ\text{C}$ to $+70^\circ\text{C}$, $V_{CC} = 5.0\text{V} \pm 10\%$ (Note 1)

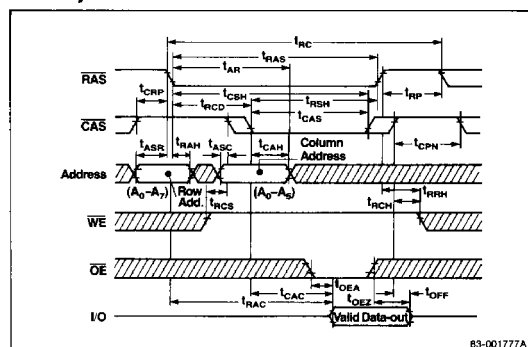
Parameter	Symbol	Limits						Unit	Test Conditions
		μ PD41416-12		μ PD41416-15		μ PD41416-20			
		Min	Max	Min	Max	Min	Max		
Data-in hold time	t_{DH}	35		45		55		ns	(Note 14)
Data-in hold time referenced to $\overline{\text{RAS}}$	t_{DHR}	95		120		155		ns	
Refresh period	t_{REF}		2		2		2	ms	
$\overline{\text{WE}}$ command setup time	t_{WCS}	0		0		0		ns	
$\overline{\text{CAS}}$ to $\overline{\text{WE}}$ delay	t_{CWD}	95		120		155		ns	
$\overline{\text{RAS}}$ to $\overline{\text{WE}}$ delay	t_{RWD}	155		195		255		ns	
Access time from $\overline{\text{OE}}$	t_{OEA}	30		40		50		ns	
Data delay time	t_{OED}	30		40		50		ns	
$\overline{\text{OE}}$ command hold time	t_{OEH}	0		0		0		ns	
Output turn-off delay from $\overline{\text{OE}}$	t_{OEZ}	0	30	0	40	0	50	ns	(Note 10)

Notes:

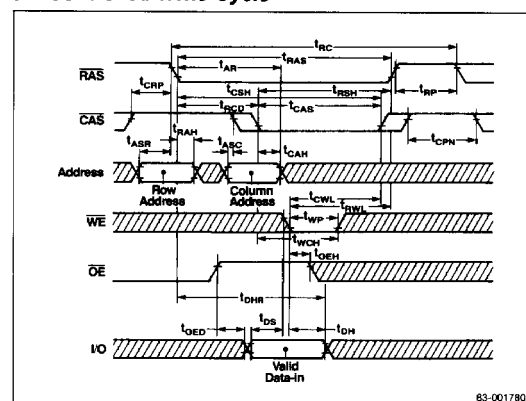
- (1) All voltages referenced to GND.
- (2) An initial pause of $100\mu\text{s}$ is required after power-up followed by any 8 $\overline{\text{RAS}}$ cycles before proper device operation is achieved.
- (3) AC measurements assume $t_T = 5\text{ns}$.
- (4) V_{IH} (min) and V_{IL} (max) are reference levels for measuring timing of input signals. Transition times are measured between V_{IH} and V_{IL} .
- (5) I_{CC1} , I_{CC3} and I_{CC4} depend on output loading and cycle rates. Specified values are obtained with the outputs open.
- (6) The minimum specifications are used only to indicate cycle time at which proper operation over the full temperature range ($T_A = 0^\circ\text{C}$ to 70°C) is assured.
- (7) Load = 2 TTL loads and 100pF .
- (8) Assumes that $t_{RCD} \leq t_{RCD}(\text{max})$. If t_{RCD} is greater than the maximum recommended value shown in this table, t_{RAC} will increase by the amount that t_{RCD} exceeds the value shown.
- (9) Assumes that $t_{RCD} \geq t_{RCD}(\text{max})$.
- (10) $t_{OFF}(\text{max})$ and $t_{OEZ}(\text{max})$ define the time at which the output achieves the open circuit condition and are not referenced to V_{OH} or V_{OL} .
- (11) Operation within the $t_{RCD}(\text{max})$ limit insures that $t_{RAC}(\text{max})$ can be met. $t_{RCD}(\text{max})$ is specified as a reference point only. If t_{RCD} is greater than the specified $t_{RCD}(\text{max})$ limit, then access time is controlled exclusively by t_{CAC} .
- (12) t_{CRP} requirement should be applicable for $\overline{\text{RAS}}$, $\overline{\text{CAS}}$ cycles preceded by any cycle.
- (13) Either t_{RRH} or t_{RCH} must be satisfied for a read cycle.
- (14) These parameters are referenced to $\overline{\text{CAS}}$ leading edge in early write cycles and to $\overline{\text{WE}}$ leading edge in delayed write or read-modify-write cycles.

Timing Waveforms

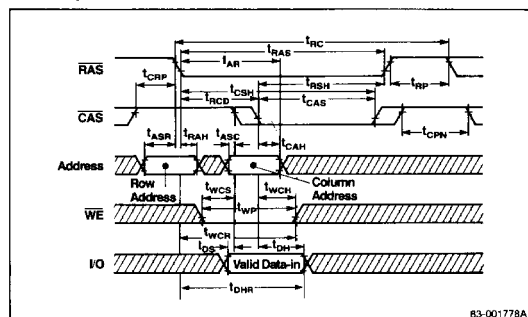
Read Cycle



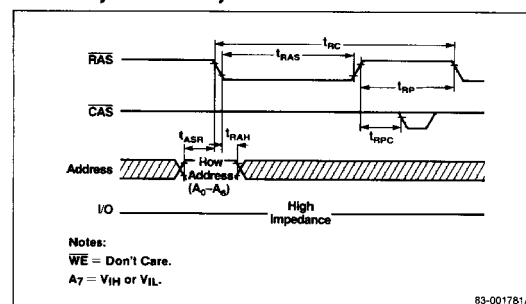
$\overline{OE}$ -Controlled Write Cycle



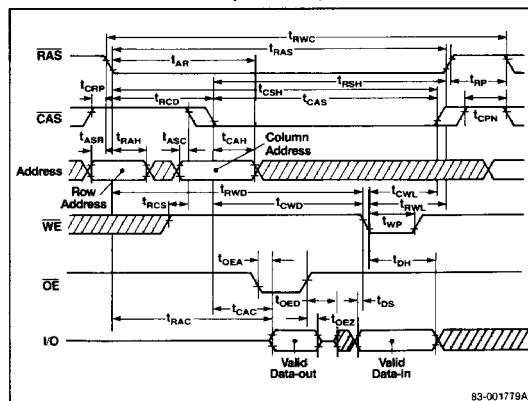
Write Cycle (Early Write)



"RAS-Only" Refresh Cycle



Read-Write / Read-Modify-Write Cycle



Hidden Refresh Cycle

